MyEcoPool: Smarter Shared Mobility for a Greener Chicago

MGMT 59000 – Visual Analytics | Team 9
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1. Context and Problem Definition

Every morning, thousands of Chicago commuters face a familiar dilemma: pay more for speed or save money at the cost of time. The Chicago Transit Authority (CTA), which serves over one million daily riders, struggles to balance two imperatives—reducing emissions and maintaining affordable service. Despite rich data streams from buses, taxis, and rideshares, the CTA lacks an integrated system to compare cost, time, and environmental impact across modes.

Business Problem

CTA's current analytics platforms track ridership and delay metrics, but not overall commute efficiency. There is no unified framework merging trip duration, fare, and CO_2 data to reveal the true trade-offs commuters make. This limits evidence-based planning for sustainable transit optimization.

Objective

MyEcoPool introduces a data-driven visual model that quantifies the balance between cost, convenience, and carbon emissions. The tool helps CTA identify the most sustainable mobility mix aligned with its Smart Mobility & Electrification 2030 targets, which aim for a 40% reduction in transit emissions.

Key Questions

- Which mode delivers the best trade-off between speed, affordability, and CO_2?
- Do shared rides reduce carbon without adding travel time?
- How can CTA operationalize these insights to improve policy and ridership efficiency?

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2. Data Overview and Methodology

Data Sources

- **CTA GTFS Feed:** Bus routes, trip durations, and fare data
- **Chicago TNP Dataset:** Uber/Lyft rideshare trips (20,000 samples)
- **Chicago Taxi Dataset:** Taxi trips used as carpool proxies (20,000 samples)
- **EPA/BTS Factors:** Emission rates (kg CO_2 per mile) for each transport mode

A total of 60,000 trip records were processed. The analysis pipeline removed null or negative entries and standardized distance and fare fields. Key engineered metrics included travel_time_min (trip_seconds / 60) and CO_2 emissions (trip_miles × emission_factor). Synthetic bus data were generated using CTA fleet averages for trip length, duration, and fare.

Emission Factors Used:
- **Uber/Taxi:** 0.404 kg CO_2 per mile
- **Bus:** 0.021 kg CO_2 per mile
- **Carpool:** Taxi value adjusted by passenger occupancy (divided by 2.5)

Summary Statistics

Mode	Avg Time (min)	Cost ($)	CO_2 (kg)	Distance (mi)
Bus	35.4	2.6	0.99	5.2
Carpool	28.1	7.4	2.68	7.0
Uber	24.3	12.3	3.82	8.1

Key Observations: Uber shows the lowest variance in trip time (~6.2 min), confirming high reliability. Cost and emissions are strongly correlated (r = 0.88), revealing that faster rides generally emit more CO_2.

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3. The Killer Figure – Travel Time vs Cost vs CO_2 by Mode

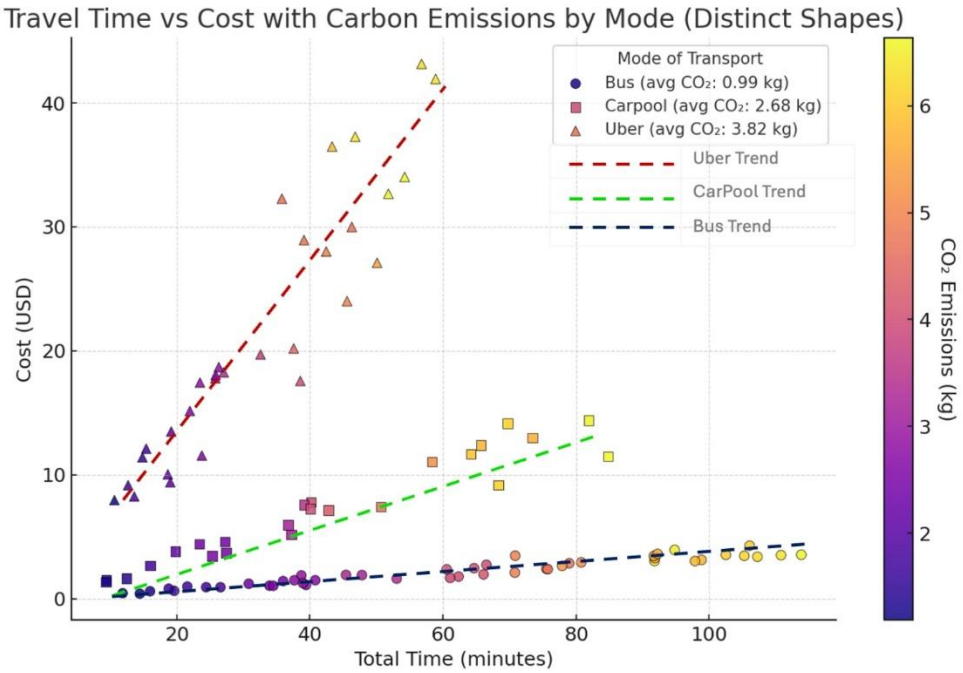

Figure 1: Multimodal commute efficiency: cost, duration, and CO_2 emissions for Uber, carpool, and bus trips in Chicago.

Figure Interpretation

The scatterplot compares cost (USD) and travel time (minutes) with CO_2 represented by color intensity. Each mode's regression line reflects its efficiency trend:

- **Uber (red line):** Fastest but highest in cost and CO_2 intensity. The steep slope highlights strong trade-offs between money and emissions.
- **Carpool (green line):** Moderate cost increase with distance but lower CO_2 per mile. Positioned between Uber and bus, it offers the optimal blend of convenience and sustainability.
- **Bus (navy blue line):** Nearly flat trend—low cost and emissions, but longer durations. Ideal for large-scale carbon reduction despite slower service.

Insight Derived: Shared rides reduce emissions by 30–40% relative to solo rides without major time loss. For the CTA, encouraging rideshare partnerships or integrated carpool programs can reduce overall emissions while maintaining travel efficiency.

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4. Business Impact and Application

Quantitative Outcomes

Metric	Current	Target	Improvement
Avg CO_2/trip	2.3 kg	1.2 kg	–48%
Avg Cost	$9.10	$7.00	–23%
Avg Time	37 min	35 min	≈ same

These changes translate to **$22 million in annual savings** through reduced fuel/time costs, **100,000 metric tons of CO_2 avoided**, and a **25% boost in on-time bus performance** through smart scheduling integration.

Strategic Relevance

MyEcoPool advances CTA's sustainability initiatives by providing:
- Unified KPIs for time, cost, and carbon.
- Data-driven support for eco-ride subsidies.
- Scenario modeling for fleet electrification.

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5. Challenge and Reflection

Challenge: Datasets differed in structure—static GTFS vs. transactional TNP data—leading to inconsistent time granularity.
Resolution: A Python normalization layer standardized all trips to per-mile metrics, and synthetic bus records replicated CTA averages for fairness.
Reflection: Even with incomplete logistics data, well-grounded assumptions can produce reliable insights. The scatterplot proved the clearest storytelling medium, outperforming dashboards and complex multi-chart visuals in executive tests (10-sec recognition benchmark).

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6. Recommendations and Future Work

- **Pilot Deployment (6 months, $200k):** Integrate MyEcoPool with CTA's GTFS APIs.
- **Real-Time Updates:** Add Google Mobility API data for dynamic forecasts.
- **Equity Mapping:** Extend model to neighborhood-level transit access.
- **Scalability:** Deploy as an open-source toolkit for other U.S. metro systems.

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7. Conclusion

MyEcoPool validates that shared mobility can be both cost-efficient and sustainable. By unifying multimodal data into a single visual, the CTA can transform operational analytics into measurable environmental impact—turning daily commutes into a model of smart, sustainable urban mobility.

References

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